

Chance Constraint Optimization

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Summer 2025

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Introduction

Optimization in an Uncertain World

- How can a robot minimize its energy consumption while navigating, given that its sensor readings are uncertain and it must still avoid obstacles with high probability?
- How do we manage a power grid when wind/solar energy is unpredictable?
- Chance Constrained Optimization (CCO) can be seen as a framework for handling uncertainty in optimization problems.



CCO Real-World Applications

- **Energy Management:** Ensuring power supply meets demand with high probability, given the uncertainty of renewable sources like wind and solar.
- **Finance:** Designing investment portfolios where the probability of losing more than a certain amount is kept low.
- **Robotics & Autonomous Systems:** Path planning where a robot must avoid obstacles with high confidence despite noisy sensors.
- **Supply Chain Management:** Managing inventory to meet uncertain customer demand without frequent stockouts.

Preliminaries

Chance Constrained Optimization

Definition (Chance Constrained Optimization Problem [KHVR20])

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & f(x) \\ \text{s.t.} \quad & \mathbb{P}\{g(x, Z) \leq 0\} \geq 1 - \delta \end{aligned} \tag{1}$$

where $\mathcal{X} \subset \mathbb{R}^d$ is a (deterministic) closed convex set,, $Z \sim P_Z$ is random vector in $\mathbb{R}^m$, $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is our *objective function*, $g : \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}$ our *chance function* and $1 - \delta$ our *safety probability level*, typically close to 1.

- We assume f and g to be *convex functions* w.r.t. x [ROC70].
- Not only is Chance Constraint optimization an NP-Hard problem, but *even finding local optima is also NP-Hard*.

Bi-level reformulation

First, we reformulate the chance constraint using the notion of $(1 - \delta)$ -quantile.

Definition (Quantile)

Let Y be a scalar random variable, $F_Y(t)$ its Cumulative Distribution Function and $\delta \in [0, 1]$ our level. Y 's $(1 - \delta)$ -quantile [Hab96] denoted $Q_{1-\delta}(Y)$ is defined as:

$$Q_{1-\delta}(Y) = \inf\{t \in \mathbb{R} : F_Y(t) \geq 1 - \delta\}. \quad (2)$$

Intuitively, it can be interpreted as the inverse of the CDF¹ of a random variable, i.e. $F_Y^{-1}(1 - \delta)$.

¹CDF stands for Cumulative Distribution Function, which describes the probability that a random variable takes a value less than or equal to a given point.

Quantile reformulation

We can rewrite our previously defined chance constraint using quantiles [LMA21]:

$$\mathbb{P}\{g(x, Z) \leq 0\} \geq 1 - \delta \iff Q_{1-\delta}(g(x, Z)) \leq 0. \quad (3)$$

Intuitively, this means that the left $1 - \delta$ proportion of the mass of $g(x, Z)$'s density should lie below 0. Using this equivalence, it follows that our main (Problem 1) can be rewritten as *Bi-level problem*:

$$\begin{cases} \min_{x \in \mathcal{X}} & f(x) \\ \text{s.t.} & \mathbb{P}\{g(x, Z) \leq 0\} \geq 1 - \delta \end{cases} \iff \begin{cases} \min_{x \in \mathcal{X}} & f(x) \\ \text{s.t.} & s \leq 0 \\ & s = Q_{1-\delta}(g(x, Z)) \end{cases}. \quad (4)$$

Challenge

$s = Q_{1-\delta}(g(x, Z))$ is a non-convex, **non-smooth constraint**.

Superquantile definition

Definition (Superquantile a.k.a. CVaR)

Let Y be a random variable, $\delta \in [0, 1]$ our level and $Q_{1-\delta}(Y)$ our quantile at the level $1 - \delta$. Y 's $(1 - \delta)$ -superquantile, denoted $SQ_{1-\delta}(Y)$ is defined as:

$$SQ_{1-\delta}(Y) = \mathbb{E}\{Y \mid Y \geq Q_{1-\delta}(Y)\} \quad (5)$$

which can be equivalently defined as (from [ROC70])

$$SQ_{1-\delta}(Y) = \frac{1}{\delta} \int_{1-\delta}^1 Q_p(Y) dp. \quad (6)$$

Superquantile definition

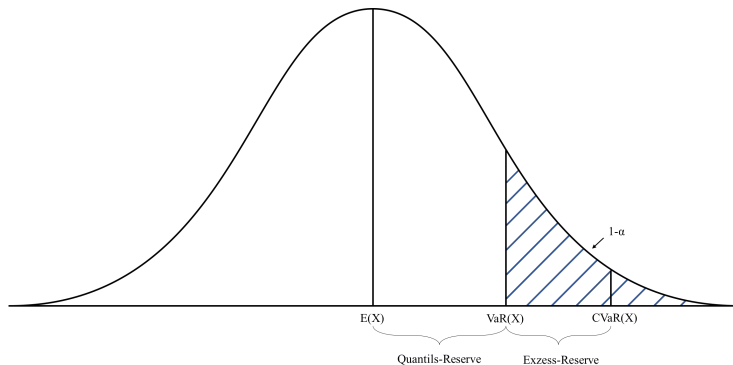


Figure: Superquantile can be intuitively viewed as the expected value of all outcomes greater than a certain level.

Variational formulation for Superquantile

Definition

Using (6), one can introduce the following function, involving the auxiliary variable s .

$$G_\delta(s) := s + \frac{1}{\delta} \mathbb{E}[\max\{Y - s, 0\}]$$

Function G is convex and continuously differentiable w.r.t s [ROC70].

Variational formulation for Superquantile

Lemma

For a given random variable Y and a level δ , superquantile can be determined by following formula [ROC70].

$$SQ_{1-\delta}(Y) = \min_{s \in \mathbb{R}} G_{\delta}(s). \quad (7)$$

Moreover, the left endpoint of the solution set of (7) is $Q_{1-\delta}(Y)$.

Bi-level reformulation

Definition

$$G_{\delta}(x, s) := s + \frac{1}{\delta} \mathbb{E}_Z[\max\{g(x, Z) - s, 0\}]$$

CCO problem can be rewritten as follows:

$$\left\{ \begin{array}{ll} \min_{x \in \mathcal{X}} & f(x) \\ \text{s.t.} & s \leq 0 \\ & s = Q_{1-\delta}(g(x, Z)) \end{array} \right. \longleftrightarrow \left\{ \begin{array}{ll} \min_{x \in \mathcal{X}} & f(x) \\ \text{s.t.} & s \leq 0 \\ & s \in S(x) = \arg \min_{s \in \mathbb{R}} G_{\delta}(x, s) \end{array} \right. . \quad (8)$$

Bi-level reformulation

Lemma

If $P_Z \ll \text{Leb}$, i.e., P_Z is absolutely continuous and $g(x, \cdot)$ is a continuous function for every $x \in \mathcal{X}$, then

$$S(x) = \arg \min_{s \in \mathbb{R}} s + \frac{1}{\delta} \mathbb{E}[\max\{g(x, Z) - s, 0\}],$$

is a singleton.

- We will denote $s^*(x)$, leading to the following reformulation of the CCO:

$$\min_{x \in \mathcal{X}} f(x) \text{ s.t. } s^*(x) \leq 0. \quad (9)$$

Algorithm

Augmented Lagrangian method

- Write (9) by its mini-max reformulation:

$$\min_{x \in \mathcal{X}} \max_{\lambda \geq 0} f(x) + \lambda s^*(x). \quad (10)$$

However, $\max_{\lambda \geq 0} f(x) + \lambda s^*(x)$ is *non-smooth* w.r.t. x .

- To address this, we use the augmented Lagrangian:

$$\min_{x \in \mathcal{X}} \max_{\lambda \geq 0} P(x, \lambda) = \min_{x \in \mathcal{X}} \max_{\lambda \geq 0} \left[f(x) + \lambda s^*(x) - \frac{\mu}{2} \lambda^2 \right]. \quad (11)$$

Here, the added μ -regularization term $-\frac{\mu}{2} \lambda^2$ serves to soften the effect of the λ -penalized constraint $\lambda s^*(x)$.

Augmented Lagrangian method

By solving maximization over λ , we relax (Problem 9) as the simpler:

$$\min_{x \in \mathcal{X}} F(x) = \min_{x \in \mathcal{X}} f(x) + \frac{s^*(x)}{2} \left[\frac{s^*(x)}{\mu} \right]_+ \quad (12)$$

with $F(x)$ being our *objective function*, $s^*(x)$ denoting $\arg \min_{s \in \mathbb{R}} G_\delta(x, s)$, $[\cdot]_+$ the $\max(\cdot, 0)$ or $\text{ReLU}(\cdot)$ function.

$s^*(x)$ is not differentiable!

Theorem (Danskin)

Let $X \subseteq \mathbb{R}^n$ be open and $S \subseteq \mathbb{R}^m$ be a convex, compact set. Consider a function $F : X \times S \rightarrow \mathbb{R}$ such that:

- 1 $F(x, s)$ is $\mathcal{C}^1$ in x ,
- 2 $F(x, s)$ is $\mathcal{C}^2$ in s ,
- 3 For each $x \in X$, the problem $\min_{s \in S} F(x, s)$ has a **unique** minimizer $s^*(x)$,
- 4 $\nabla_s^2 F(x, s)$ is positive definite at $s = s^*(x)$.

Then the solution map $x \mapsto s^*(x)$ is continuously differentiable ($\mathcal{C}^1$).

- $G_\delta(x, s) = s + \delta^{-1} \mathbb{E}[\max\{g(x, Z) - s, 0\}]$ is $\mathcal{C}^1$ function but **not necessarily a $\mathcal{C}^2$ one!**
- We therefore approximate $G_\delta(x, s)$ with a $\mathcal{C}^2$, strongly convex surrogate $\hat{G}_{\delta, \theta, \epsilon}(x, s)$.

Approximation for G_δ

In order to make $G_\delta(x, \cdot)$ **twice differentiable**, we approximate $\max\{x, 0\}$ with a $\mathcal{C}^2$ function $h_\theta(x)$ in the definition of $G_\delta(x, s)$.

$$h_\theta(x) = \begin{cases} -\frac{x^4}{16\theta^3} + \frac{3x^2}{8\theta} + \frac{x}{2} + \frac{3\theta}{16} & \text{if } x \in [-\theta, \theta] \\ \max\{x, 0\} & \text{otherwise} \end{cases} \quad (13)$$

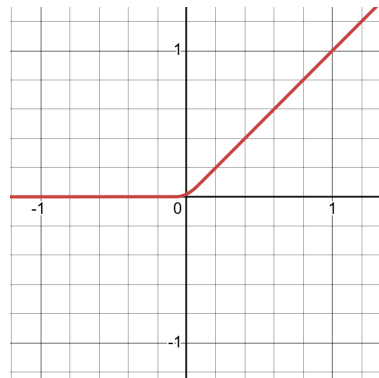


Figure: h_θ graph with $\theta = 0.1$.

Approximation for G_δ

- Add a square term in s to $G_\delta(x, s)$:

$$\hat{G}_{\delta, \theta, \epsilon}(x, s) := s + \frac{1}{\delta} \mathbb{E}_Z[h_\theta(g(x, Z) - s)] + \frac{\epsilon}{2} s^2. \quad (14)$$

- By Danskin theorem, $\hat{s}_{\theta, \epsilon}^*(x) := \arg \min_s \hat{G}_{\delta, \theta, \epsilon}(x, s)$ is $\mathcal{C}^1$.

Minimization of $\hat{F}$

We express and compute

- We relax $F(x) := f(x) + \frac{s^*(x)}{2} \left[\frac{s^*(x)}{\mu} \right]_+$ to the following differentiable objective:

$$\hat{F}(x) := f(x) + \frac{\hat{s}_{\theta,\epsilon}^*(x)}{2} \left[\frac{\hat{s}_{\theta,\epsilon}^*(x)}{\mu} \right]_+.$$

- To minimize $\hat{F}$, we will first use the *gradient descent algorithm*.
Need to compute the partial derivatives $\frac{\partial}{\partial x} \hat{F}(x)$ and $\frac{\partial}{\partial x} \hat{s}_{\theta,\epsilon}^*(x)$.
- $\nabla \hat{F}(x) = \nabla_x f(x) + \nabla_x \hat{s}_{\theta,\epsilon}^*(x) \left[\frac{\hat{s}_{\theta,\epsilon}^*(x)}{\mu} \right]_+$
- $\nabla_x \hat{s}_{\theta,\epsilon}^*(x) = -\partial_{xs}^2 \hat{G}_{\delta,\theta,\epsilon}(x, s)|_{s=\hat{s}_{\theta,\epsilon}^*(x)} \left[\partial_s^2 \hat{G}_{\delta,\theta,\epsilon}(x, s)|_{s=\hat{s}_{\theta,\epsilon}^*(x)} \right]^{-1}$

Gradient computation

$$\begin{aligned}
 \frac{\partial G_{\delta,\theta,\epsilon}(x, s_{\theta,\epsilon}^*(x))}{\partial s} = 0 &\implies \frac{d}{dx} \left[\frac{\partial G_{\delta,\theta,\epsilon}(x, s_{\theta,\epsilon}^*(x))}{\partial s} \right] = 0 \\
 &\iff \frac{\partial^2}{\partial x \partial s} G_{\delta,\theta,\epsilon}(x, s_{\theta,\epsilon}^*(x)) + \frac{\partial}{\partial x} s_{\theta,\epsilon}^*(x) \frac{\partial^2}{\partial s_{\theta,\epsilon}^2} G_{\delta,\theta,\epsilon}(x, s_{\theta,\epsilon}^*(x)) = 0 \\
 &\iff \frac{\partial}{\partial x} s_{\theta,\epsilon}^*(x) = - \frac{\partial^2}{\partial x \partial s} G_{\delta,\theta,\epsilon}(x, s_{\theta,\epsilon}^*(x)) \left[\frac{\partial^2}{\partial s_{\theta,\epsilon}^2} G_{\delta,\theta,\epsilon}(x, s_{\theta,\epsilon}^*(x)) \right]^{-1}.
 \end{aligned}
 \tag{15}$$

Plug-in estimator of $\hat{G}_{\delta,\theta,\epsilon}(x, s)$

- Access to P_Z through the samples $\{z_j\}_{j=1}^M$. $\hat{G}_{\delta,\theta,\epsilon}(x, s)$ needs to be estimated.
- Plug-in estimator of $\hat{G}_{\delta,\theta,\epsilon}(x, s)$:

$$\hat{G}_{\delta,\theta,\epsilon}(x_k, \hat{s}_k; \{z_j\}_{j=1}^M) := s + \frac{1}{\delta M} \sum_{j=1}^M h_{\theta}(g(x, z_j) - s) + \epsilon \cdot \frac{s^2}{2} \quad (16)$$

- Then

$$\hat{\nabla} s_{\theta,\epsilon}^*(x, s) = \partial_{xs}^2 \hat{G}_{\delta,\theta,\epsilon}(x, s; \{z_j\}_{j=1}^M) \left[\partial_s^2 \hat{G}_{\delta,\theta,\epsilon}(x, s; \{z_j\}_{j=1}^M) \right]^{-1},$$

$$\nabla_x \hat{F}(x, s) := \nabla_x f(x) - \hat{\nabla} s_{\theta,\epsilon}^*(x, s) \left[\frac{\hat{\nabla} s_{\theta,\epsilon}^*(x, s)}{\mu} \right]_+.$$

Algorithm Hyper-gradient Augmented method

Require: ϵ, θ , learning rates $\eta, \gamma > 0$, number of iterations N, N' .

- 1: Sample a batch of M points $\{z_j\}_{j=1}^M$ from the distribution P_Z .
 - 2: Initialize x_0, s_0
 - 3: **for** $k = 0, 1, \dots, N' - 1$ **do**
 - 4: Initialize $\hat{s}_k = s_k$
 - 5: **for** $k' = 0, 1, \dots, N - 1$ **do**
 - 6: $\hat{s}_{k+1} \leftarrow \hat{s}_{k'} - \gamma_{k'} \nabla_s \hat{G}_{\delta, \theta, \epsilon}(x_k, \hat{s}_{k'}; \{z_j\}_{j=1}^M)$
 - 7: **end for**
 - 8: $s_{k+1} = \hat{s}_N$
 - 9: $\hat{x}_{k+1} \leftarrow \hat{x}_k - \eta_k \nabla_x \hat{F}(x_k, s_{k+1})$
 - 10: **end for**
 - 11: **return** $\hat{x}'_N$
-

Experiments

Example 1: Definition

First Example is from Conformal Predictive Programming paper [Zha+24], a simple uni-variate problem.

$$\begin{aligned}f(x) &= x^3 e^x \\g(x, z) &= 50ze^x - 5 \\Z &\sim \text{Exp}\left(\frac{1}{3}\right)\end{aligned}\tag{17}$$

with safety probability level $1 - \delta = 0.90$ and 500 samples.

Example 1: Learning curve

- Smooth trajectory
- Fast convergence
- Low variance

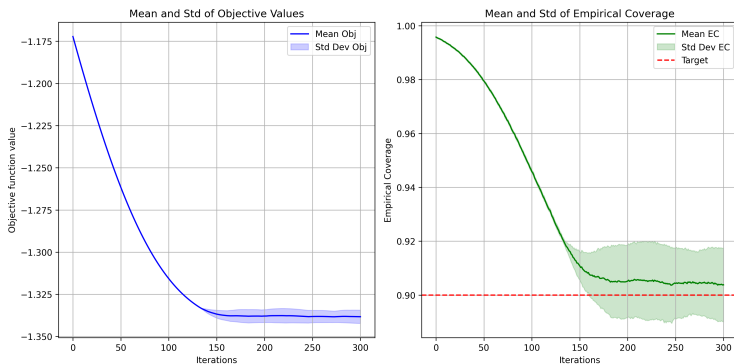


Figure: Objective and Empirical coverage plot over 100 runs.

Example 1: Comparison

Algorithm	Objective Mean	Objective Std	EC Mean	Run Times Mean
Ours	-1.3386	0.0028	0.9032	2.8345
CPP-MIP	-1.3398	0.0034	0.8986	0.5917
CPP-KKT	-1.3396	0.0038	0.8982	35.3135

Table: Algorithm performance over 100 runs

Example 2: Definition

Example for demonstrating the performance of algorithms in the high-dimensional case.

$$\begin{aligned}f(x) &= ||x - 2 \times \mathbf{1}||^2 \\g(x, z) &= Z^T x - 1 \\Z &\sim \mathcal{N}(\mathbf{0}, I_{d \times d})\end{aligned}\tag{18}$$

with safety probability level $1 - \delta = 0.95$, 100 samples and dimension equal to 10.

Example 2: Learning Curve

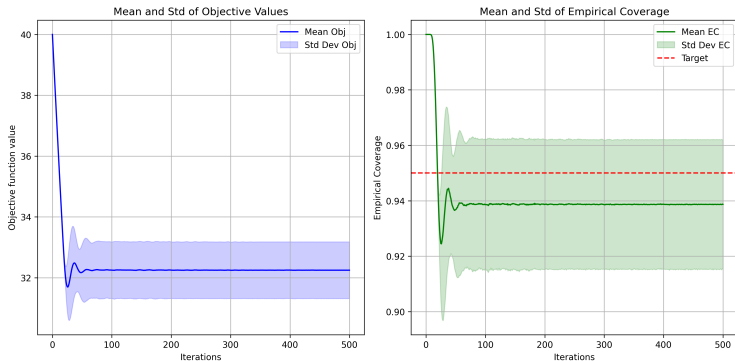


Figure: Objective and Empirical coverage plot over 30 runs.

Example 2: Comparison

Algorithm	Objective Mean	Objective Std	EC Mean	Run Times Mean
Ours	31.8462	0.96066	0.9281	3.6852
CPP-MIP	29.8731	1.1626	0.8375	4.6654
CPP-KKT	29.9085	0.9959	0.8457	154.6919
TACO	36.9806	0.8045	0.9977	3.5867

Table: Algorithm performance over 20 runs

- results showing vulnerability of Conformal Prediction algorithms against dimension and sample number.

Example 3: Definition

30 dimensional version of the second example:

$$\begin{aligned}f(x) &= \|x - 2 \times \mathbf{1}\|^2 \\g(x, z) &= Z^T x - 1 \\Z &\sim \mathcal{N}(\mathbf{0}, I_{d \times d})\end{aligned}\tag{19}$$

with safety probability level $1 - \delta = 0.95$ and 100 samples.

Example 3: Learning Curve

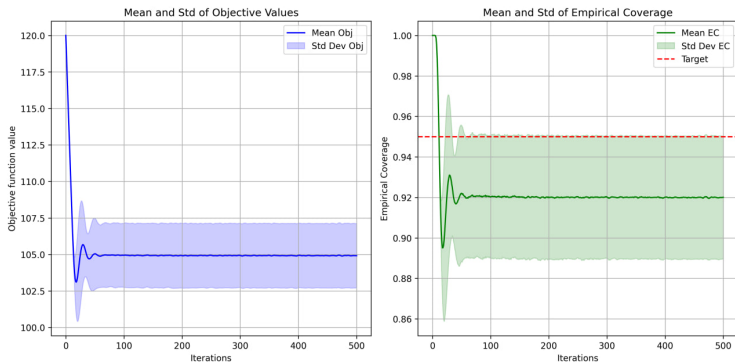


Figure: Objective and Empirical coverage plot over 30 runs.

Example 3: Comparison

Algorithm	Objective Mean	Objective Std	EC Mean	Run Times Mean
Ours	105.2801	2.0875	0.9249	7.0542
CPP-MIP	87.9880	7.1422	0.6753	82.1295

Table: Algorithm performance over 30 runs

Example 4: Definition

250 sample version of the second example:

$$\begin{aligned}f(x) &= \|x - 2 \times \mathbf{1}\|^2 \\g(x, z) &= Z^T x - 1 \\Z &\sim \mathcal{N}(\mathbf{0}, I_{d \times d})\end{aligned}\tag{20}$$

with safety probability level $1 - \delta = 0.95$ and dimension equal to 10.

Example 4: Learning Curve

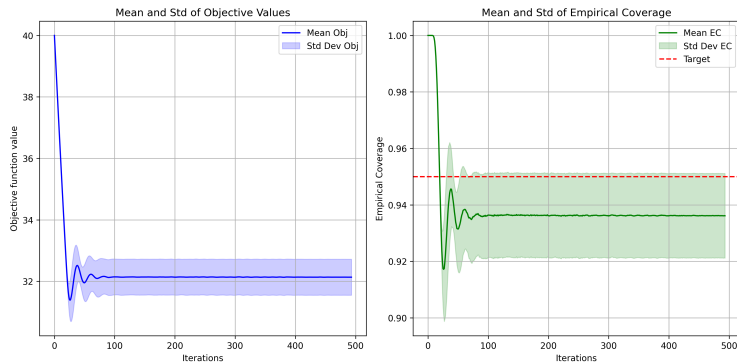


Figure: Objective and Empirical coverage plot over 30 runs.

Example 4: Comparison

Algorithm	Objective Mean	Objective Std	EC Mean	Run Times Mean
Ours	31.9731	0.7483	0.9318	4.0398
CPP-MIP	31.1557	0.5767	0.8918	216.2729

Table: Algorithm performance over 30 runs

Future Works

Robust image classification under random corruptions

Critical Issue

Deep learning models achieve exceptional performance on clean benchmark datasets, but frequently fail in real-world applications where input quality is compromised.

Why This Matters

- Real-world images are rarely perfect
- Environmental factors introduce unavoidable corruptions
- Safety-critical applications demand reliability

Real-World Scenarios

Autonomous driving (rain, fog) • Medical imaging (noise, artifacts)
Surveillance (motion blur, compression) • Mobile photography (low light, shake)

CCO framework

- Corrupted training set
- Clean test set
- $f_{\theta}(x) = [p_1, \dots, p_k]$
- p_i probability of belonging x to class i
- $C(x) = \max_i f_{\theta}(x)_i$
- $m_{\theta}(x) = f_{\theta}(x)_{C(x)} - \max_{i \neq C(x)} f_{\theta}(x)_i$

$$\begin{aligned}
 \min_{x \in \mathcal{X}} \quad & \text{CrossEntropyLoss}(f_{\theta}(x), y) \\
 \text{s.t.} \quad & \mathbb{P}(m_{\theta}(x) \geq c) \geq 1 - \delta
 \end{aligned} \tag{21}$$

Thank You!

References

- [Hab96] Shelby J. Haberman. "Quantiles". In: *Advanced Statistics: Description of Populations*. New York, NY: Springer New York, 1996, pp. 367–404. ISBN: 978-1-4757-4417-0. DOI: 10.1007/978-1-4757-4417-0_7. URL: https://doi.org/10.1007/978-1-4757-4417-0_7.
- [KHVR20] Willem K. Klein Haneveld, Maarten H. van der Vlerk, and Ward Romeijnders. "Chance Constraints". In: *Stochastic Programming: Modeling Decision Problems Under Uncertainty*. Cham: Springer International Publishing, 2020, pp. 115–138. ISBN: 978-3-030-29219-5. DOI: 10.1007/978-3-030-29219-5_5. URL: https://doi.org/10.1007/978-3-030-29219-5_5.
- [LMA21] Yassine Laguel, Jérôme Malick, and Wim Ackooij. *Chance constrained problems: a bilevel convex optimization perspective*. 2021. arXiv: 2103.10832 [math.OC]. URL: <https://arxiv.org/abs/2103.10832>.
- [ROC70] R. TYRRELL ROCKAFELLAR. "Convex Functions". In: *Convex Analysis*. Princeton University Press, 1970, pp. 23–31. ISBN: 9780691015866. URL: <http://www.jstor.org/stable/j.ctt14bs1ff.8> (visited on 04/12/2025).
- [Zha+24] Yiqi Zhao et al. *Conformal Predictive Programming for Chance Constrained Optimization*. 2024. arXiv: 2402.07407 [eess.SY]. URL: <https://arxiv.org/abs/2402.07407>.